

Vivek Jagadeesh

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EDUCATION

Worcester Polytechnic Institute (WPI)

Worcester, MA

Master of Science in Computer Science (Thesis Track), GPA: 4.0/4.0

Graduation May 2027

- Research Advisor: Professor Robert J. Walls

Bachelor of Science in Computer Science – Concentration in Cybersecurity, GPA: 3.97/4.0

Graduated May 2026

SKILLS

Cybersecurity & Systems: Systems Security, Linux Kernel Development, Compiler-based Defenses, Cryptography

Programming Languages: C, C++, Rust, Assembly (x86 & ARM), Python

Development Tools: LLVM/Clang Tooling, QEMU, BusyBox, Docker, VirtualBox, VMware

Cloud & Databases: AWS, Azure, PostgreSQL, MongoDB

AI & Machine Learning: Scikit-learn, AWS Bedrock, RAG, OpenSearch

EXPERIENCE

Systems Security Research Intern

May 2026 – Aug 2026

MIT Lincoln Laboratory: Group 53

Lexington, MA

- Built a Clang-based transpiler to instrument Hardware Assisted Kernel Compartmentalization in the Linux kernel.
- Designed a two-pass AST pipeline using cross-AST imports and targeted transformations for minimal patches.
- Created a regression test suite to compare output against expected patches for instrumentation correctness.
- Instrumented PAC and MTE primitives in kernel source to provide hardware-backed compartment isolation.
- Compartmentalized IPv6 and nftables loadable kernel modules and validated isolation with BusyBox in QEMU.

Software Engineering Intern

Jun 2025 – Aug 2025

Waters Corporation

Milford, MA

- Engineered a serverless RAG platform on AWS (Lambda, Bedrock) with real-time UI streaming.
- Built automated AI Agent order processing flows and presented demos to leadership.

RESEARCH & PROJECTS

Secure Memory Allocator for the Linux Kernel, WPI Cake Lab Research Project

Aug 2025 – Present

- Designed and implemented a low-overhead Linux kernel memory allocator to mitigate stale-reference exploits.
- Researched temporal memory safety vulnerabilities in the Linux kernel and analyzed exploitation techniques.
- Implemented the allocator in the Linux kernel and deployed it on an embedded target with six protected objects.
- Evaluated allocator performance against existing solutions using targeted macro- and microbenchmarks.
- Submitted a paper detailing novel techniques to a top-tier security conference.
- Won WPI's Provost's MQP Award for the top undergraduate research project in Computer Science.

Smart Grid Security: PSEM Interception Simulation, CS 4404: Networks Security

Jan 2025 – Mar 2025

- Developed a client-server ANSI C12.18 PSEM simulator on dual Arduino Mega 2560 microcontrollers.
- Implemented PSEM packet generation and parsing to emulate ANSI C12.19 smart-meter table communications.
- Captured and analyzed raw UART traffic with a logic analyzer to demonstrate protocol-level packet interception.
- Recovered plaintext credentials from intercepted C12.19 password-table transmissions in protocol.
- Demonstrated how intercepted credentials could enable unauthorized authenticated meter write requests.

LEADERSHIP & TEACHING EXPERIENCE

Student Teaching Assistant

May 2024 – May 2026

Worcester Polytechnic Institute

Worcester, MA

- Mentored 24 students across 6 agile teams in design, development, and cloud deployment of web applications.
- Led recitation sessions on cloud deployment and containerization using AWS EC2, RDS, and Docker.
- Conducted weekly code reviews and provided actionable feedback to enforce software engineering standards.
- Guided teams in implementing agile ceremonies (Scrum, retrospectives, backlog refinement).